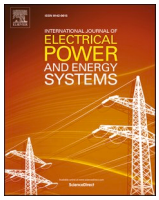
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Improved P&O MPPT algorithm with efficient open-circuit voltage estimation for two-stage grid-integrated PV system under realistic solar radiation

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ABSTRACT

Considering the vast improvement of photovoltaics (PVs) efficiency, this paper proposes a robust modified perturb and observe (MPO) maximum power point tracking (MPPT) algorithm. The control strategy of the proposed MPO-MPPT algorithm is based on dividing the solar module/cell P-V curve into four operating areas depending on the open-circuit voltage estimation method. The two areas, which are located far from the maximum power point (MPP), utilize large voltage fixed step-size that improves the tracking speed of the PV system. Otherwise, a small step-size P&O MPPT algorithm is used to minimize the steady-state oscillations for the other two areas due to the close locations of these areas to the MPP. As well, the proposed tracking algorithm eliminates the necessity of wide small step-size iterations by concentrating small step-size search area to 15% of the whole P-V operation region. To prove the effectiveness of the proposed MPO algorithm compared with the conventional MPPT techniques; sinusoidal, ramp, and one-day (10 hr.) irradiance profiles are applied to the solar PV system. The proposed MPO-MPPT algorithm based solar PV system has been built using MATLAB/SIMULINK software. Obviously, the system results confirm the theoretical analysis of the proposed algorithm, which boosts the PV system tracking efficiency to 99.7%.

1. Introduction

Recently, many governments worldwide, led by USA and China, have vowed to eliminate greenhouse gas emissions to achieve a low-carbon energy future. Renewable energy based distributed energy resources (DERs) are key enabling technology to accomplish the transition to a carbon-free energy sector [1]. Among the existing DERs, solar photovoltaics (PVs) are the freely, abundant, and most promising energy resources that supplies the humankind with large amount of energy [2-4]. As well, PV generation systems are characterized by scalability, pollution free, absence of fuel cost, and low maintenance requirement. However, PV energy generation confronts several issues such as; high installation cost, low-efficiency, and weather-dependency energy generation [5-8]. Considering the effect of different weather conditions, typical solar system efficiency is within 9% and 17% [9]. Moreover, the weather dependency of solar energy makes solar systems unstable power supplier. Hence, efficient control strategies are essential to ensure the PV systems efficient and secure operation.

The solar cell I-V and P-V characteristic curves are nonlinear and depend on the solar insolation and module temperature as depicted in Fig. 1. Obviously, a unique point exists on the P-V curve that termed as the maximum power point (MPP) at which optimal power is extracted from the solar cell. Thus, different tracking algorithms have been presented for maximum power point tracking (MPPT) of the solar PV applications at different irradiance conditions [10-14]. Solar MPPT algorithms have been broadly categorized into two major categories; conventional and soft computing MPPT techniques. Conventional MPPT techniques are commonly used in many applications due to the implementation simplicity and low cost such as; fractional short-circuit current (FSCC), fractional open-circuit voltage (FOCV), Hill climbing, perturb and observe, and hill climbing strategies [15-17]. In advance requirements of system data for the FSCC and FOCV limits their applications due to their module dependency operation. In spite of the popularity of HC, P&O, and IC MPPT strategies and their efficient operation under uniform irradiation, they fail to track the MPPT under rapidly changing insolation and partial shading conditions. In addition, these techniques suffer from the low tracking speed and high steady state oscillations that forms the main shortcomings of these methods. On the other side, soft computing based MPPT techniques such as; fuzzy

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Nomenclature

V_{MPP}	PV voltage at MPP
V_{oc}	Open circuit voltage of PV array
I_{ph}	Photo-generated current
I_{sat}	Reverse saturation current
I_{sc}	PV short circuit current
G	Global insolation
T_{eff}	Tracking efficiency
BC	Boost converter
MPPT	Maximum Power Point Tracking
CPO	Conventional step-size P&O
MPO	Modified P&O
IC	Incremental Conductance
LS-PO	Large-step P&O
FLC	Fuzzy Logic Control
SMC	Sliding Mode Control
ANN	Artificial Neural Network
PSO	Particle swarm Optimization
ANFIS	Adaptive Neuro Fuzzy Interface System
ESC	Extremum seeking control
CS	Cuckoo-Search
GWO	Gray wolf optimization
STC	Standard Test Condition
SS-PO	Small-step P&O
DER	Distributed Energy Resources
FSCC	Fractional Short Circuit Current
FOCV	Fractional Open Circuit Voltage

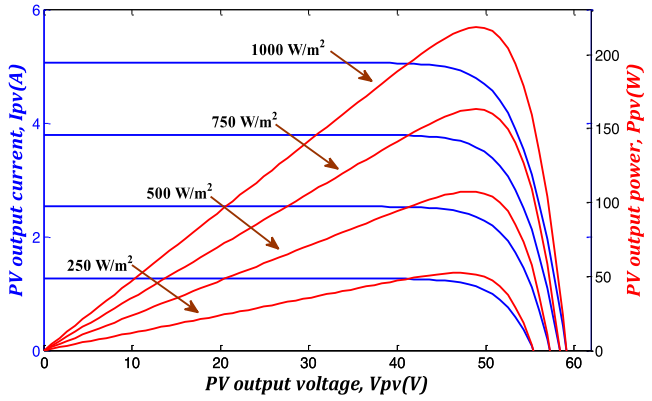


Fig. 1. Solar cell output characteristics under different levels of irradiance.

logic control (FLC), sliding mode control (SMC), artificial neural network (ANN) and metaheuristic-based algorithms (i.e.; particle swarm optimization and genetic algorithm) were developed to overcome the common associated drawbacks of conventional techniques [18-20]. These computational-based MPPT algorithms are capable of handling the system nonlinearity to track the global optimal MPP, especially under partial shading operating conditions. However, the computation burden and complexity are the great concerns with soft computing MPPT control algorithms [5,21,22]. Many efforts have been exerted by researchers on developing these MPPT algorithms to improve solar PV systems response. In [23], an improved P&O MPPT algorithm has been introduced to enhance the tracking efficiency of PV system under fast changing irradiation condition. However, it shows a slow tracking response of the PV system with high steady-state system oscillations. In [17], an improved IC MPPT technique has been presented to improve the system dynamic response by considering acceptable error-level under fast rapidly varying irradiation condition. However, the presented technique executes number of calculations that requires some special controller specifications as well as slowing the system response. In [24], a new MPPT algorithm is presented considering IC tracker to improve the system dynamic response. However, it still has a relatively slow time response compared with many algorithms. In [19], a novel MPPT technique has been presented considering photodiode as the irradiance sensor. Thus, it can estimate the level of irradiation in problematic scenarios. However, the presented method is applicable for small and simple PV systems. In [25], a novel MPPT technique is presented with an accurate MPPT dynamic response and tracking strategy. Despite the improved response of the presented topology in [25], it has a

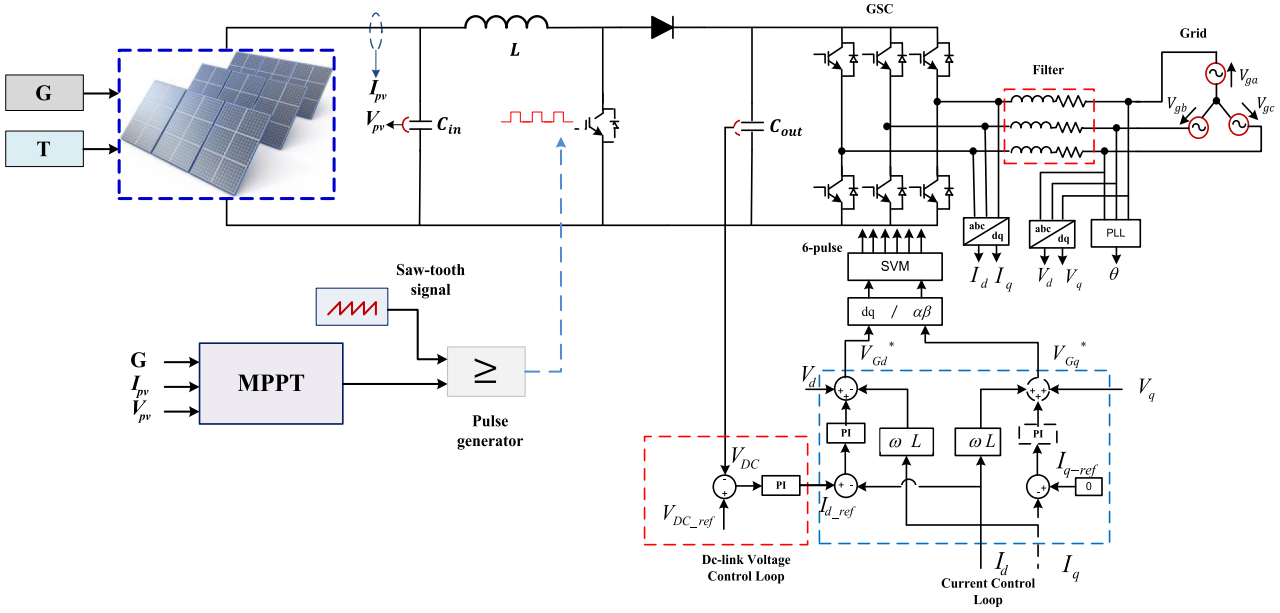


Fig. 2. Proposed two-stage grid-integrated PV system.

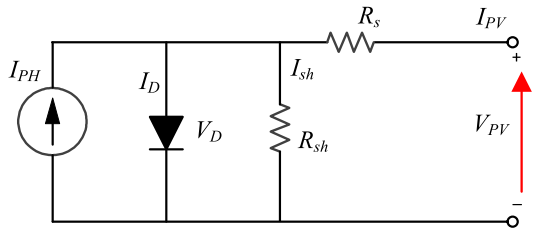


Fig. 3. PV cell model.

Table 1
Canadian CS5P-220 M PV module characteristics.

Maximum power, P_{MPP}	220 (W)
Voltage at MPP, V_{MPP}	48.3159 (V)
Current at MPP, I_{MPP}	4.54758 (A)
Open circuit voltage, V_{OC}	59.2618 (V)
Short circuit current, I_{SC}	5.09261 (A)
Temperature, ate STC	25 C

comparatively low MPPT speed. Thus, a neural network based MPPT algorithm is presented in [26] considering two dual hidden layers. However, it exhibits a complex computational burden for MPPT strategy implementation. In [15], an improved P&O MPPT technique is presented considering dynamic step-size for oscillation minimization. In [27], an improved MPPT algorithm based particle-swarm optimization adaptive neuro fuzzy interface system (PSO-ANFIS) is presented. Despite the efficient operation of the presented algorithm, it requires high controller specifications. In addition, extremum seeking control (ESC) and improved ripple-based ESC have been presented in [28] and [29], respectively. An improved MPPT based scanning technique is introduced in [30] for MPP extraction. Also, a new hybrid MPPT method is proposed in [31] based on direct perturbation strategy. However, these MPPT control strategies requires additional sensors for hardware implementation. In [32], Cuckoo-Search (CS) based MPPT algorithm is proposed for efficiency improvement and enhanced convergence time. However, it requires higher controller specifications for hardware implementation. In [33], an improved MPPT technique for partial shading condition is presented considered gray wolf optimization (GWO) algorithm. In spite of the improved dynamic behavior of the tracker, it suffers from the implementation complexity.

Motivated by the aforementioned literature survey, this paper proposes an efficient multi-sector P&O MPPT algorithm integrated with solar PV system considering an efficient open-circuit voltage estimation strategy in order to reduce the required number of sensors, as illustrated in Fig. 2. In addition, the proposed technique divides the PV operating region into four operating areas, in which large voltage step-size is applied for the two areas located far from the MPP to improve the tracking speed of the proposed technique. Otherwise, small voltage step-size is applied to minimize the steady-state oscillations of the PV system. The proposed tracking algorithm has been applied for different realistic solar radiations, which confirms the features of the proposed tracking algorithm under efficient operation of open-circuit voltage estimation strategy. In addition, the proposed controller boosts the tracking efficiency of the PV system to 99.7%. The rest of the paper has been structured as follows; Section 2(A and B) illustrate the mathematical modeling of the PV system and open circuit voltage estimation method, respectively. The boost converter modelling is illustrated and analyzed in Section 3. Section 4 discusses the proposed MPO operating strategy. The three-phase grid-tie inverter is modelled in section 5. Section 6 depicts the two-stage MPPT results considering an accurate open-circuit voltage estimator with detailed analysis under different realistic solar irradiance. Finally, Section 7 concludes the paper results.

2. System modelling

2.1. PV cell model

The solar PV cell is the main basic item in solar generation systems that converts solar photo-energy into electrical energy for customer's usage. Solar PV cell are accumulated through parallel and series combination to boost the output voltage, current, and the required rating of the solar array. Different solar cell models have been presented, however, single diode model is the common cell model due its simplicity [1]. Hence, the solar cell equivalent circuit based on single diode model is depicted in Fig. 3. The PV voltage-current relation can be described as follows;

$$I = N_p I_{ph} - N_p I_{sat}\left[e^{\frac{q\left(\frac{v_{pv}}{N_s}+\frac{R_s \bullet i_{pv}}{N_p}\right)}{A \bullet k \bullet T}} - 1\right] - \frac{N_p \bullet \frac{v_{pv}}{N_s} + R_s \bullet i_{pv}}{R_{sh}} \tag{1}$$

where; i_{pv} is the output current of the PV array (A), v_{pv} is the terminal voltage of the PV array (V), N_s and N_p are the number of series and parallel connected PV modules, I_{ph} is the cell photo-generated current (A), I_{sat} is the reverse saturation current (A), and R_s - R_{sh} are the series and shunt module equivalent resistances, respectively. *K, q, A,* and *T* are Boltzmann constant (1.38 * 10^{-23} J/K), electron charge (1.6022 * 10^{-19}C), p-n ideality factor, and PV cell temperature, respectively.

The photo-generated current (photo-current) is related to the applied solar irradiance and temperature on the PV array, and the PV array characteristics. Thus, the photo-current formula can be expressed as follows [1,15];

$$I_{ph} = \frac{G}{1000}(I_{sc} + K_i(T - T_i)) \tag{2}$$

where; I_{sc} is the short circuit current at standard temperature and irradiance condition ($G = 1000\ W/m^2$ *and* $T_i = 298\ K$), K_i is the short circuit current temperature coefficient and T is the PV cell reference temperature. Also, the PV cell reverse saturation current varies with the cell temperature based on the following expression [1,3];

$$I_{sat} = \frac{I_{sc} + K_i(T - T_i)}{\exp\left(\frac{V_{oc}+K_v(T-T_i)}{V_t}\right) - 1} \tag{3}$$

where; V_{oc} is the open-circuit voltage of the PV cell, K_v is the open-circuit voltage coefficient and V_t is the thermal voltage ($V_t = K.T/q$).

In addition, the PV module short-circuit current and open-circuit voltage formulas can be expressed, in terms of the array solar irradiation and its temperature, as follows [15,22];

$$\mathrm{I_{sc}} = \frac{\mathrm{G}}{1000}[\mathrm{I_{sc(STC)}} + \mu_{\mathrm{I_{sc}}}(T - \mathrm{T_{(STC)}})] \tag{4}$$

$$V_{oc} = V_{oc(STC)} + \mu_{V_{oc}}(T - T_{(STC)}) \tag{5}$$

where; I_{sc} and $I_{sc(STC)}$ are the short-circuit currents of the solar PV module at actual and standard-test-conditions. V_{oc} and $V_{oc(STC)}$ are the open-circuit voltages of solar PV module at actual and standard-test-conditions. T and $T_{(STC)}$ are the temperature of the PV modules (K) at actual and STC (*25 °C or 298.15 K*). Also, μ_{Vov} and μ_{Isc} are the solar cell thermal coefficient for open-circuit voltage ($V/^oC$) and short-circuit current ($A/^oC$), respectively.

In addition, the diode current (I_D) can be formulated as follows;

$$I_D = I_{sat} \bullet \left[e^{\left(\frac{V_D}{N_s \bullet V_t}\right)} - 1\right] \tag{6}$$

where; V_D is voltage drop across the diode.

Solar PV array based on Canadian solar module CS5P-220 M PV is used for the proposed system verification, where the module parameters

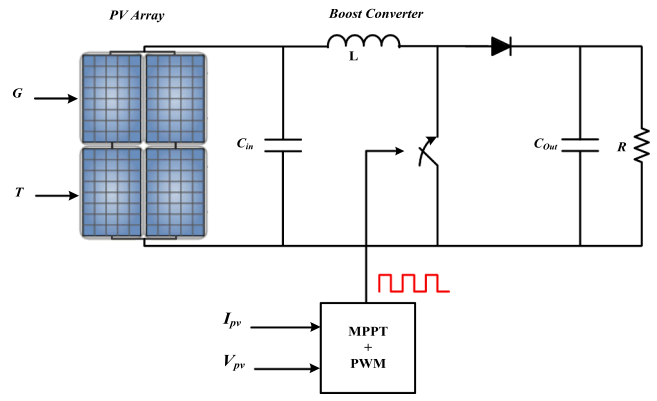


Fig. 4. Block Diagram of PV array Connected to BC.

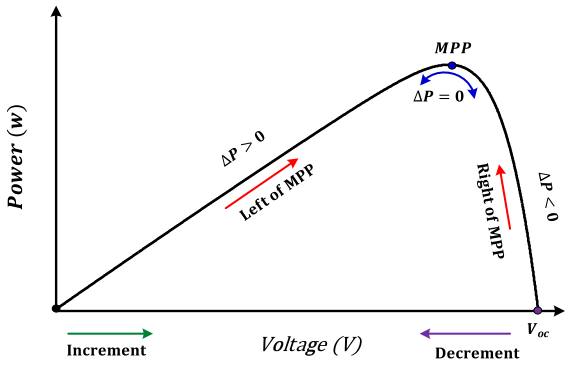


Fig. 5. Conventional P&O MPPT operation strategy.

at STC are listed in Table 1.

2.2. Open circuit voltage estimation strategy

Based-on (5), the open circuit voltage can be formulated, in terms of PV module temperature, as follows;

$$V_{oc} = V_{oc(STC)} + k_t\left(T - T_{(STC)}\right) \bullet V_{oc(STC)} \tag{7}$$

where; k_t is the open-circuit voltage thermal coefficient.

The open-circuit voltage formula in (7) is expressed in terms of the cell temperature, however, it depends on the solar irradiance on the solar cell. Therefore, the open-circuit voltage can be expressed using (6). During open-circuit test, V_D and I_D are equal to I_{ph} and V_{oc}, respectively. Thus, the photo-generated current can be formulated as follows;

$$I_{ph} = I_{sat} \bullet \left[e^{\left(\frac{V_{oc}}{N_s \bullet V_t}\right)} - 1\right] \tag{8}$$

Hence, the open-circuit voltage can be simplified as follows;

$$V_{oc} = N_s \bullet V_t \bullet \left[ln\left(\frac{I_{ph} + I_{sat}}{I_{sat}}\right)\right] \tag{9}$$

Also, $I_{ph} \gg I_{sat}$, then the open-circuit voltage can be formulated as follows;

$$V_{oc} = N_s \bullet V_t \bullet \left[ln\left(\frac{I_{ph}}{I_{sat}}\right)\right] \tag{10}$$

At STC, the open-circuit voltage can be expressed as follows;

$$V_{oc,STC} = N_s \bullet V_t \bullet \left[ln\left(\frac{I_{ph,STC}}{I_{sat,STC}}\right)\right] \tag{11}$$

where; $I_{sat} = I_{sat,STC}$

Based-on (10) and (11);

$$V_{oc} - V_{oc,STC} = N_s \bullet V_t \bullet \left[ln\left(\frac{I_{ph}}{I_{sat}}\right) - ln\left(\frac{I_{ph,STC}}{I_{sat,STC}}\right)\right] \tag{12}$$

Thus;

$$V_{oc} = V_{oc,STC} + N_s \bullet V_t \bullet \left[ln\left(\frac{I_{ph}}{I_{ph,STC}}\right)\right] \tag{13}$$

Considering the effect of I_{ph} from (13) into (7);

$$V_{oc} = V_{oc,STC} + N_s \bullet V_t \bullet \left[ln\left(\frac{I_{ph}}{I_{ph,STC}}\right)\right] + k_t\left(T - T_{(STC)}\right) \bullet V_{oc(STC)} \tag{14}$$

Consequently, the open-circuit voltage of the PV array can be analytically and accurately estimated using (14). However, the photo-generated current requires a sensor, therefore short-circuit current estimation is used to calculate the photo-current as follows;

Based-on the single-diode model in Fig. 3, the shunt resistor current can be formulated as follows;

$$I_{sh} = \frac{V_D}{R_{sh}} = \frac{V_{PV} + I_{PV} \bullet R_s}{R_P} \tag{15}$$

Also, the photo-current can be expressed as follows;

$$I_{ph} = I_{PV} + I_D + I_{sh} \tag{16}$$

Substituting from (6), (15) into (16);

$$I_{ph} = I_{PV} + I_{sat} \bullet \left[e^{\left(\frac{V_{PV} + I_{PV} \bullet R_s}{N_s \bullet V_t}\right)} - 1\right] + \frac{V_{PV} + I_{PV} \bullet R_s}{R_P} \tag{17}$$

Also, the short-circuit current is approximately equal to the photo-generated current based-on [30]. Therefore, the short circuit can be formulated as follows;

$$I_{SC} = I_{PV} + I_{sat} \bullet \left[e^{\left(\frac{V_{PV} + I_{PV} \bullet R_s}{N_s \bullet V_t}\right)} - 1\right] + \frac{V_{PV} + I_{PV} \bullet R_s}{R_P} \tag{18}$$

Based-on the former open-circuit voltage and short-circuit current estimation methods, the number of required sensors is reduced, which reduces the system implementation cost.

3. Modelling of the boost converter

A boost converter (BC) is used as an interface between the PV array and grid inverter, which connects the PV system to the utility grid. The BC is used to track the MPP of the PV-array as well as DC-bus voltage-boosting of small scale PV systems for grid-integrated applications [34]. The BC output-voltage depends on its switch duty-cycle, which is decided by the MPPT algorithm as depicted in Fig. 4. Assuming the BC switching state as w, which is 0 when the switch is OFF and 1 as the switch is ON, as follows;

$$w = \begin{cases} 0, open switch \\ 1, closed switch \end{cases} \tag{19}$$

The dynamic model of the BC is formulated based-on Kirchhoff's laws, as follows [35];

$$\frac{di_{dc}}{dt} = \frac{V_{PV} - V_{dc}}{L} + w.\frac{V_{dc}}{L} \tag{20}$$

$$\frac{dV_{dc}}{dt} = \frac{i_{dc}}{C_{out}} - w.\frac{i_{dc}}{C_{out}} \tag{21}$$

where; V_{dc} is the chopper circuit terminal voltage and i_{dc} is the inverter input current.

The BC output-voltage can be expressed as a function of its input-voltage V_{pv} and converter switch duty-cycle d;

Table 2
Division of PV Characteristic curve.

	Starting (% of V_{oc})	Ending (% of V_{oc})	Total area (% of V_{oc})
Area (a)	0	70	70
Areas (b, c)	70	85	15
Area (d)	85	100	15

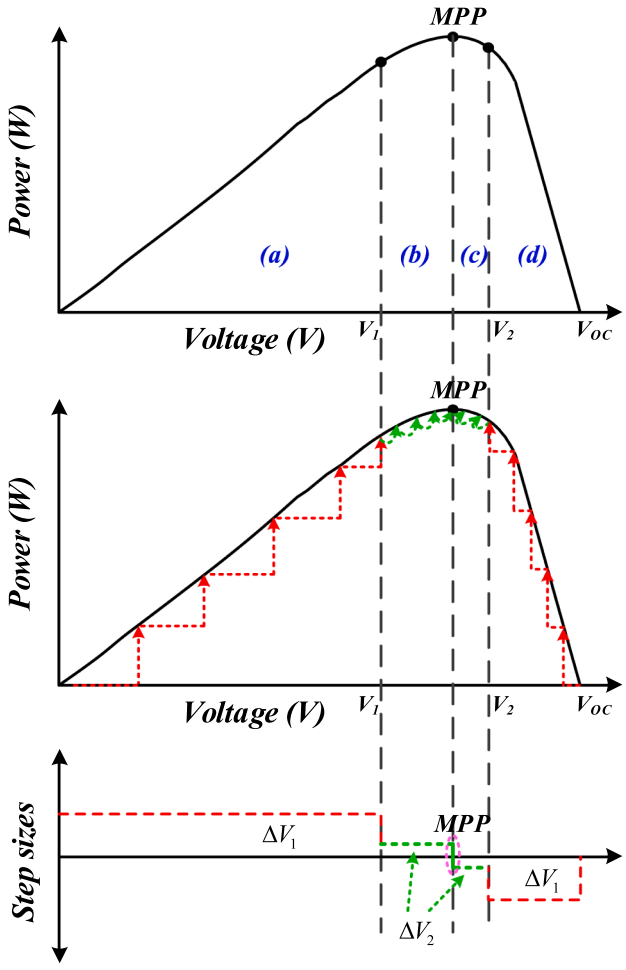


Fig. 6. The proposed MPO MPPT operation methodology.

Table 3
Scheme of the P&O algorithm.

ΔP	ΔV	Perturbation
Positive	Positive	+ve
Positive	Negative	-ve
Negative	Positive	-ve
Negative	Negative	+ve

$$\frac{V_{dc}}{V_{pv}} = \frac{1}{1-d} \tag{22}$$

Based-on equation (22), the converter output-voltage can be controlled by controlling the switch duty-cycle to track the MPP.

4. MPPT control technique

4.1. Conventional MPPT control technique

Conventional P&O (CPO) MPPT is the most common tracking strategy in many industrial applications due to many reasons such as; simplicity, low cost, reduced number of required sensors, and direct operation [1]. The operation of the CPO MPPT is illustrated in Fig. 5 and Table 2. Obviously, CPO varies the location of the system operating point by perturbing operating voltage of the PV array by voltage increment (ΔV), which disturbs the PV array output power by (ΔP). The CPO continues the operating voltage disturb until reaches the MPP location for optimal power extraction from the PV array. In addition, the CPO operation is divided into two types based on the voltage increment; a) large step-size CPO (LS-PO), which utilizes large values for operating voltage increment, and b) small step-size CPO (SS-PO) for small incremental voltage utilization. Both LS-PO and SS-PO offer advantage and suffer from some drawbacks. The LS-PO offers high MPPT speed and suffers from high peak-to-peak steady-state voltage ripples. In the other side, SS-PO minimizes the steady-state peak-to-peak voltage ripples, however, its tracking speed is very slow that diminishes the system operating efficiency.

4.2. Proposed open-circuit voltage-estimation based MPPT technique

This section illustrates the proposed modified P&O (MPO) MPPT with detailed operation analysis. The proposed MPO divides the operation region of the PV array into four operating sectors based-on the array open-circuit voltage. Thus, pre-tested open-circuit voltage is required for MPPT operation or an additional voltage sensor. However, this paper alleviates the former drawback of the MPO by using an accurate open-circuit voltage estimation method, as mathematically derived in section 3b. Thus, P-V curve of the PV array is divided into four operating sectors in-which different voltage perturbation increments can be applied based on the location of each sector. For the two sector locates far-away from the MPP, large value of (ΔV) is applied in order to fasten the MPPT speed of the PV array. Otherwise, small voltage step-size (ΔV) is applied to perturb the operating voltage due to the close location of the operating point from the MPP. The operation of the proposed MPO is illustrated in Fig. 6. Obviously, large value of (ΔV) is applied for sectors (a, and d), where small (ΔV) is used for sectors (b, and c). According to [36], the PV output voltage at MPP is about 76% of the array open-circuit voltage. Therefore, sectors' boundaries are defined based on Table 3 to ensure optimal performance of the proposed MPO under varying solar irradiation. Based on the definitions in Table 3, the two far regions (a, and d) represent 85% of the P-V curve search area in the regions ($0 < V_{pv} < V_1$) and ($V_1 < V_{pv} < V_{oc}$). During these two far regions, a large step-size is applied for MPPT of the PV system, as previously illustrated. The direction of voltage step-size perturbation can be decided based on Table 3. The duty-cycle for each operation sample can be formulated as follows;

$$\Delta d_{BC} = K \bullet \frac{\Delta P_n}{\Delta V_n} \tag{23}$$

where; K is a constant for accurate perturbation of the BC duty cycle.

Thus, the proposed MPO algorithm reduces the MPPT controller search area to 15% of the PV array open-circuit voltage. Hence, the proposed MPPT control strategy fastens the MPPT speed with reduced steady-state voltage ripples, which enhances the overall system efficiency. The flow chart of the proposed MPO MPPT algorithm is

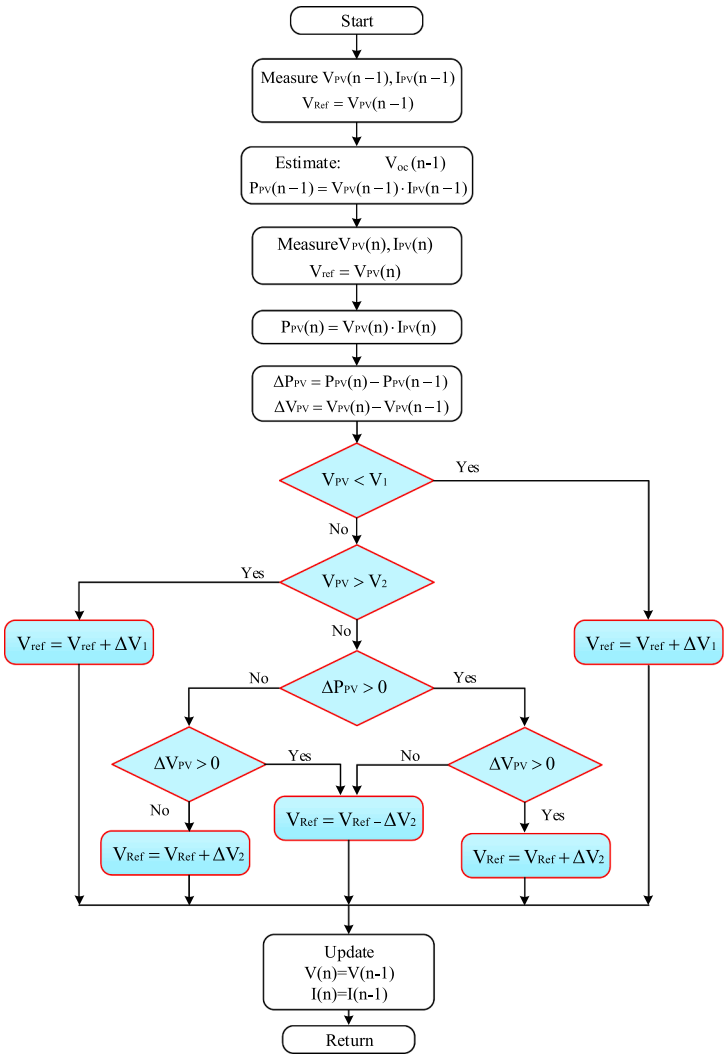


Fig. 7. Flow chart of the proposed MPO MPPT algorithm.

illustrated in Fig. 7.

5. Grid-side converter

Three-phase voltage source grid-side converter (GSC) is used to control the PV power injection into the utility grid. Hence, the inverter is used to control the grid active and reactive power as well as DC-bus voltage regulation at different climatic temperature and irradiance conditions. Also, vector-control is used to achieve unity power factor operation of the grid-integrated PV system. The DC-bus voltage is regulated via conventional PI-based outer loop, whereas the d-q axis currents are controlled through dual PI-based inner control loops. The control block diagram is illustrated in Fig. 2.

The three-phase grid voltages can be formulated as follows;

$$\begin{bmatrix} e_a \\ e_b \\ e_c \end{bmatrix} = R_f \bullet \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} + L_f \bullet \frac{d}{dt} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} + \begin{bmatrix} V_{ga} \\ V_{gb} \\ V_{gc} \end{bmatrix} \tag{24}$$

where; e_a, e_b, and e_c are the inverter output voltages, i_a, i_b, and i_c are the grid line currents, and v_{ga}, v_{gb}, and v_{gc} are the grid phase voltages.

Thus, the rotating reference-frame of the grid voltages can be expressed as follows;

$$\begin{bmatrix} v_{gd} \\ v_{gq} \end{bmatrix} = \begin{bmatrix} e_d \\ e_q \end{bmatrix} - R_f \bullet \begin{bmatrix} i_d \\ i_q \end{bmatrix} - \begin{bmatrix} \dot{\lambda}_d - \omega_0 \psi_q \\ \dot{\lambda}_q + \omega_0 \psi_d \end{bmatrix} \tag{25}$$

$$\dot{\lambda}_d = l_f \frac{d}{dt} i_d \dot{\lambda}_q = l_f \frac{d}{dt} i_q \psi_d = l_f i_d \psi_q = l_f i_q$$

where; v_{gd}, v_{gq}, e_d, and e_q are the d-axis and q-axis voltage vectors of the three-phase grid voltages, respectively. Also, ω_0 is grid fundamental frequency (Rad/s).

Also, the d-q axis representation of the active and reactive powers can be formulated as follows;

$$P = \frac{3}{2}(V_{gd} i_{d-} V_{gq} i_q) \tag{26}$$

$$Q = \frac{3}{2}(V_{gd} i_{q-} V_{gq} i_d) \tag{27}$$

6. System results and discussion

A detailed switched model of the grid-connected PV system, depicted in Fig. 2, is implemented in MATLAB/SIMULINK to assess the effectiveness of the proposed MPO algorithm. System parameters and GSC PI controllers' parameters are provided in Appendix, Tables 6 and 7, respectively. The GSC current and voltage control loops are optimally designed to provide acceptable dynamic response and ensure the overall system stability. To evaluate the performance of the proposed MPPT algorithm, the grid-connected PV system is simulated under various solar irradiance profiles such as ramp, sinusoidal and one-day irradiance

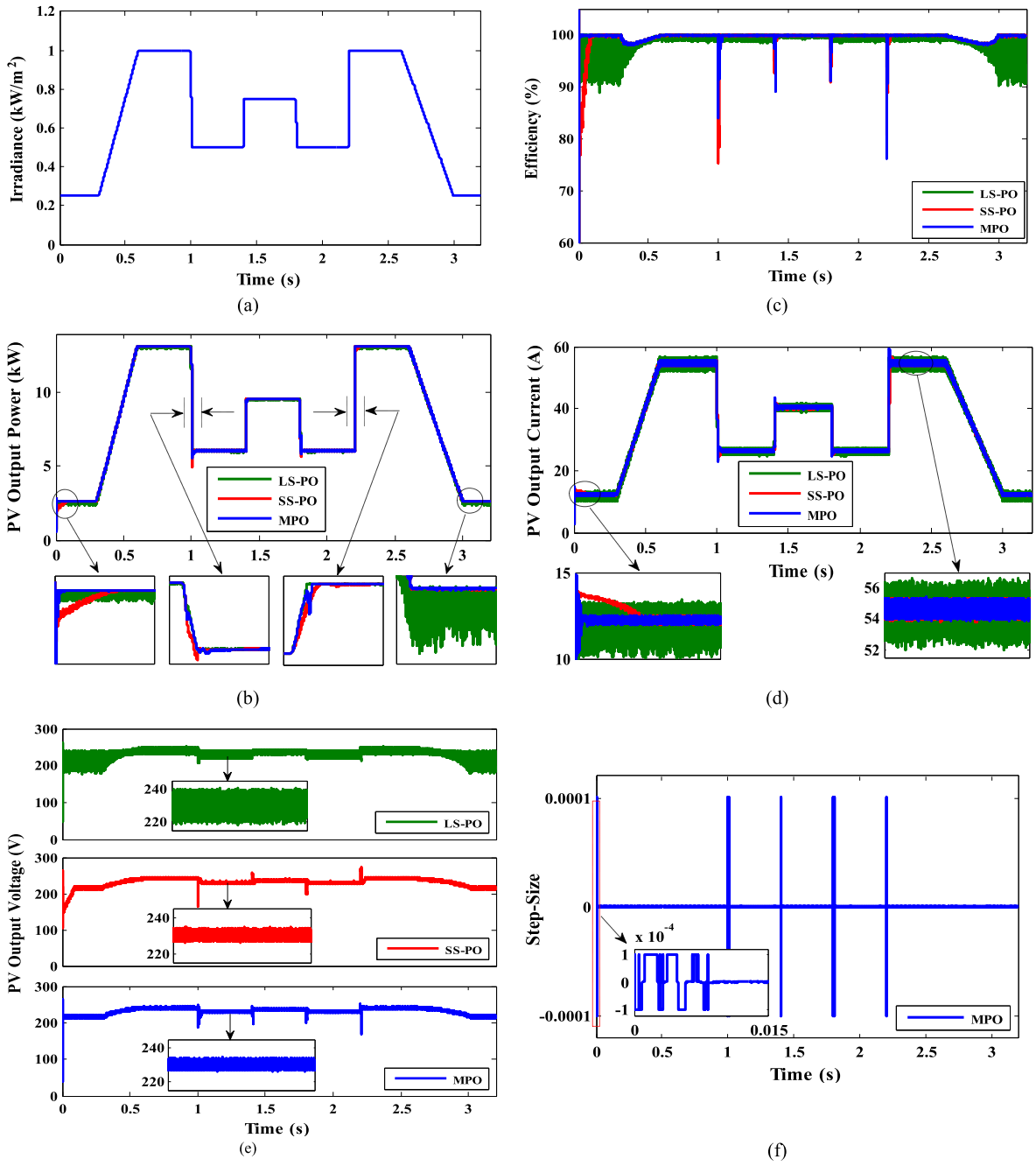


Fig. 8. PV side results; (a) Ropp irradiance profile (b) PV output-power (c) System tracking efficiency (d) PV output current (e) PV output voltage (f) Perturbation step-size.

Table 4
Comparison among various P&O MPPT algorithms at 1000 W/m^2 and 25 °C.

MPPT algorithm	Voltage ripple (V)(Peak to Peak)	ΔV	Settling time (s)	T_{eff} (%)
Conventional LS-PO	20	$\Delta V_1 = 1e-4$	0.015	98.8
Conventional SS-PO	3	$\Delta V_2 = 1e-6$	0.12	99.3
Proposed MPO	3	$\Delta V_1 = 1e-4 \Delta V_2 = 1e-6$	0.017	99.7

profiles assuming fixed ambient temperature (25 °C). Moreover, to demonstrate the merits of the proposed MPO strategy, a comparison with conventional P&O algorithms is provided. Overall, the MPO algorithm exhibits an enhanced MPPT efficiency and dynamic performance compared with the conventional algorithms.

6.1. Ropp irradiance profile [15]

To evaluate the performance of the proposed tracking algorithm under both gradual and sudden step-changes of irradiance profiles, the solar insolation profile presented in Fig. 8(a) is considered. The tracking performance of the proposed MPO algorithm, conventional LS-PO and SS-PO algorithms are compared regarding the PV output power, efficiency, and PV output current and voltage as shown in Fig. 8. Based-on

Table 5
Detailed comparison of the proposed MPO with its counterpart MPPT algorithms.

Reference	MPPT technique	Type	Required sensors	Response time (ms)	Efficiency (%)
[36]	Open-circuit voltage (OCV)	Offline/Buck converter	1 voltmeter	82	86
[37]	Short-circuit current (SCC)	Offline/Cuk converter	2 Amperemeter	80	83
[38]	P&O	Online/ boost converter	1 voltmeter 1 Amperemeter	76 (fixed step)15 (variable step)	8896
[39]	Incremental conductance	Online/	1 voltmeter 1 Amperemeter	81	95
[25]	Not reported	Online/ buck converter	1 voltmeter 1 Amperemeter	12	99.6
[18]	P&O	Online/ buck converter	1 voltmeter	50	97
	IncCond.		1 Amperemeter	60	97.8
[22]	P&O	Online/ boost converter	1 voltmeter 1 Amperemeter	Not reported	Not reported
[15]	Modified P&O	Online/ buck boost converter	1 voltmeter 1 Amperemeter	Not reported	99
[27]	PSO-ANFIS	Offline/ boost converter	1 voltmeter	10,000–17,000	97–98
	P&O-ANFIS		1 Amperemeter	12,000–19,000	85–97
	ANN			13,000–19,000	85–97
This work	Modified P&O	Online/ boost converter	Sensor-less	17	99.7

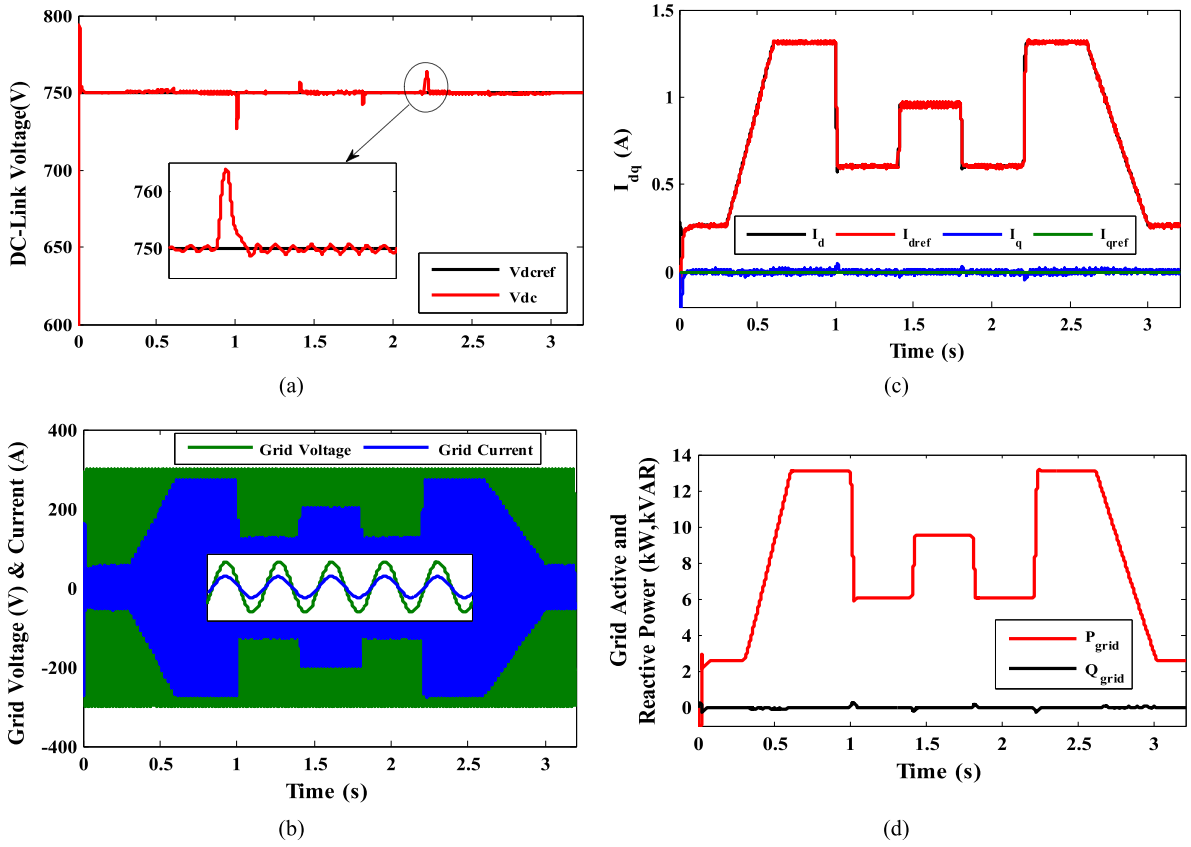


Fig. 9. Grid-side system results (a) DC-link voltage (b) Grid phase voltage and current (c) Grid d-q axis currents (d) Active and reactive power injected to the grid.

the output power depicted in Fig. 8(b), It can be observed that the proposed MPO algorithm offers a low power steady-state oscillations and small settling-time in compared with the conventional strategies. The power oscillations are almost similar to the conventional SS-PO strategy and the settling time is comparable to conventional LS-PO technique. Therefore, the proposed algorithm combines the benefits of both LS-PO and SS-PO conventional tracking algorithms. This improvement in tracking performance is reflected on the tracking efficiency as depicted in Fig. 8(c). The LS-PO algorithm has the worst tracking performance with efficiency varying in the range of 90–99.6% due to larger power oscillations, while SS-PO algorithm demonstrates improved efficiency at steady state, but its efficiency is greatly reduced at transients due to slow tracking speed. On the other hand, the proposed MPO algorithm shows enhanced tracking efficiency at transient and steady-state, which fastens the MPPT speed and diminishes steady-state oscillations.

The PV array output current and voltage are shown in Fig. 8(d) and (e), respectively. Similarly, the LS-PO attains the location of maximum power faster, however it results in larger current and voltage oscillations with a peak-to-peak voltage ripples of 30 V. In contrast, the SS-PO minimizes the steady-state voltage ripples but with degraded MPPT transient performance. The proposed MPO combines the advantageous

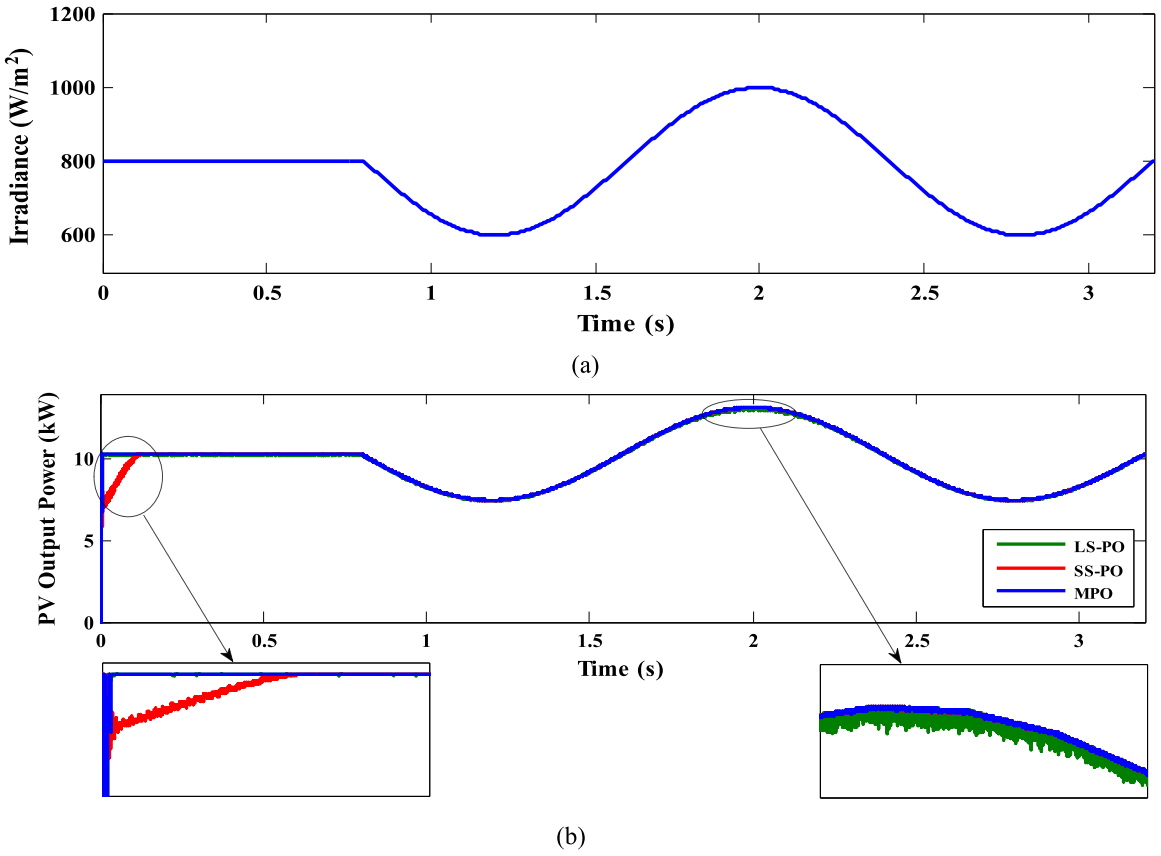


Fig. 10. (a) Sinusoidal irradiance profile (b) Comparison of the PV output power considering MPO and conventional P&O MPPT algorithms.

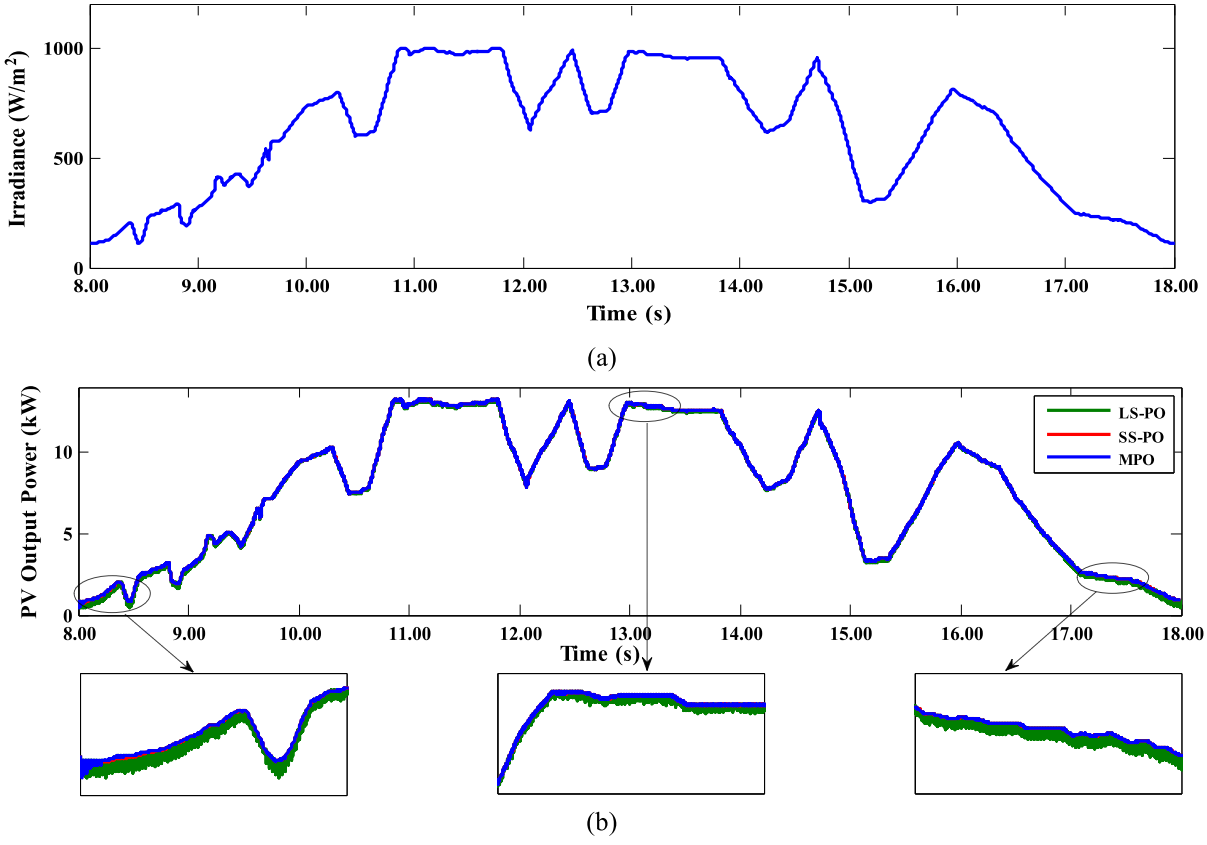


Fig. 11. System results; (a) One-day irradiance profile (b) Comparison of the PV output power of MPO and conventional P&O algorithms.

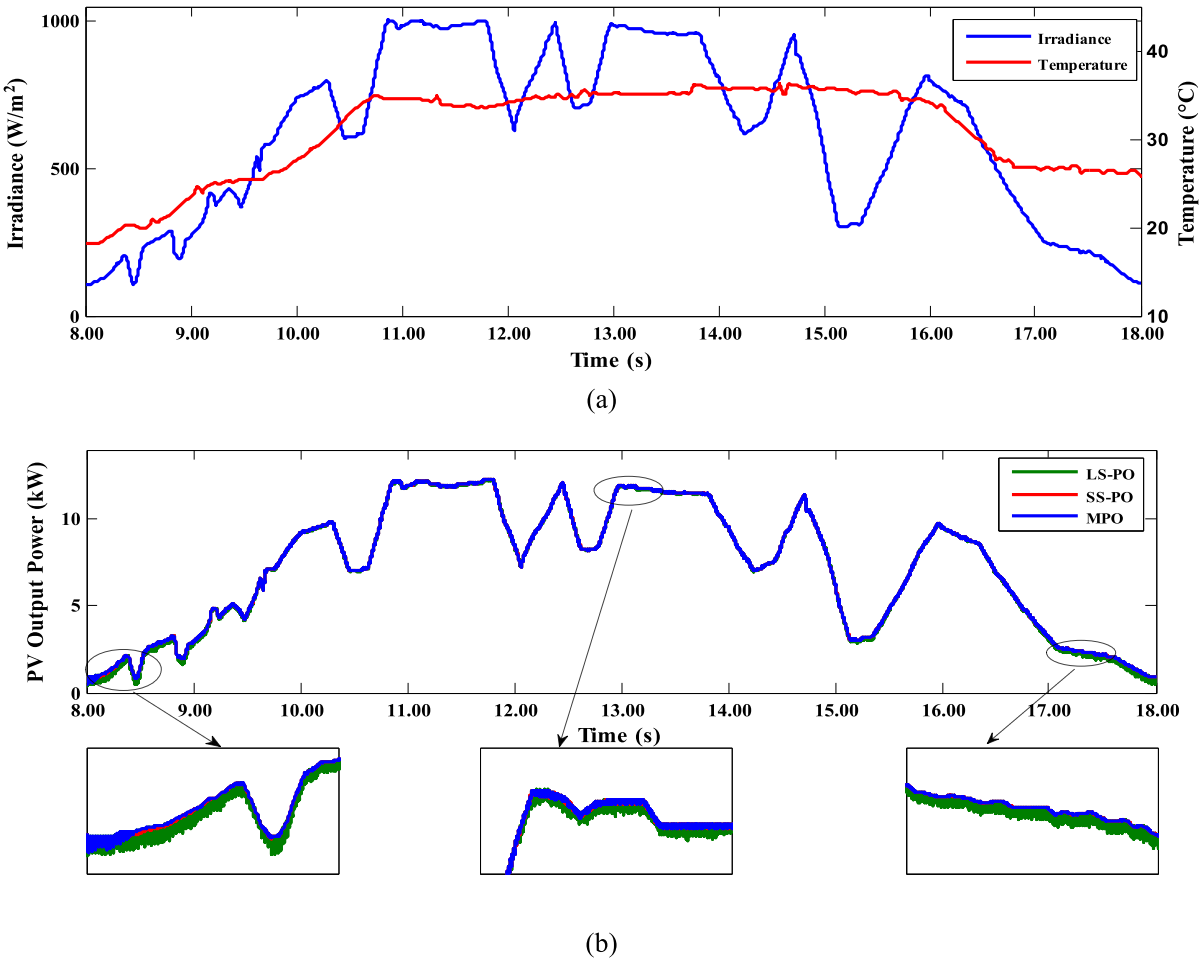


Fig. 12. System results; (a) One-day irradiance with temperature change (b) Comparison of the PV output power of MPO and conventional P&O algorithms.

features of the conventional strategies resulting in enhanced the system transient response and reduced peak-to-peak voltage ripples of 3 V. The perturbation voltage step-sizes of the proposed MPO algorithm are depicted in Fig. 8(f). The results verify the theoretical analysis for the proposed algorithm detailed operation that is provided in Section 5. It can be observed that at the beginning of the simulation, the algorithm applies large perturbation step-size and as it gets closer to the MPP as small perturbation step-size is used. The transition between large and small step sizes is determined based on the sectors defined by the open-circuit voltage of the PV array. As well, the same operational behavior can be remarked when a step change in the irradiance is applied. The algorithm initially applies a large step to enhance tracking speed and then small step is used to achieve lower steady state oscillations. A comparison of the three tracking techniques in terms of voltage ripple, settling time, and tracking efficiency is presented in Table 4. In addition, the performance of the proposed MPO algorithm has been compared with different MPPT techniques as listed in Table 5. The comparative study is based-on the online/offline efficacy, utilized DC-DC converter, number of required sensors, response time, and tracking efficiency. Obviously, the proposed MPO offers an improved operating efficiency (*99.7%*) and response time (*17 ms*) compared with the other algorithms. Moreover, the proposed MPO mitigate the requirement for voltage and current sensors, which enhances the system footprint and cost.

The performance of the GSC under the proposed MPO algorithm can be assessed using the results depicted in Fig. 9. In Fig. 9(a), the DC-link voltage is effectively regulated at its nominal value (750 V) with small oscillation level and acceptable dynamic stability under step change of the PV output-power, which demonstrates the designed controller robustness and operation reliability. In addition, the d-axis and q-axis grid current components follow their reference values as shown in Fig. 9 (b). The q-axis current is regulated to zero to achieve unity power factor operation. In Fig. 9(d), it can be noted that there is no reactive power being injected to the grid, while the active power is changing according to the output of the PV array. The grid phase voltage and current are shown in Fig. 9(d). As demonstrated by the zoomed regions, the grid voltage and current are in-phase verifying unity power factor operation of the GSC.

6.2. *Sinusoidal irradiance profile*

Despite the dynamic features of the Ropp irradiance profile, the irradiance slope is linear and that does not always represent the actual environmental conditions. Therefore, a sinusoidal irradiance profile is applied to test the proposed algorithm performance under nonlinear slope varying irradiance rate at every sampling time. The sinusoidal irradiance profile, as depicted in Fig. 10(a), begins at 800 W/m^2 and then exhibits sinusoidal changes at t = 0.8 s, peaking at 1000 W/m^2 and dipping at 600 W/m^2. The tracking performance of the proposed MPO and conventional techniques can be determined by inspecting the output power as shown in Fig. 10(b). It can be observed that the MPO algorithm demonstrates enhanced overall performance compared with conventional P&O-MPPT techniques. On one hand, the MPO exhibits a fast-tracking speed comparable to LS-PO, and on the other side, it offers low steady-state power oscillations similar to the SS-PO, that can be clearly seen by the zoomed areas of Fig. 10(b).

6.3. One-day irradiance profile [15]

To further assess the performance of the proposed MPO algorithm, a daily irradiance profile of ten hours shown in Fig. 11(a) is considered. The profile represents a real irradiance data of a typical day with intermittent clouds and occasional showers. The irradiance profile begins at 8:00 AM, which increases linearly from the morning until midday. Then it remains almost constant during afternoon, and finally drops linearly by the evening period as clearly portrayed in Fig. 11(a). A comparison of the PV output power for the proposed and conventional tracking algorithms is depicted in Fig. 11(b). Similar to the Ropp and Sinusoidal irradiance test conditions, the MPO-MPPT algorithm shows a better tracking performance compared with the conventional techniques by incorporating the LS-PO and SS-PO tracking merits considering low steady-state oscillations and fast-tracking speed. Therefore, implementing the proposed method in grid-connect PV plants would enhance the tracking performance, reduce the required number of sensors by using the proposed open-circuit voltage estimation strategy, increase the overall efficiency of the system, and reduce components stress due to lower power oscillations.

To assess the proposed algorithm performance under the variation of both solar irradiance and temperature, the one-day irradiance profile with temperature variation throughout the day is considered as shown in Fig. 12(a) [15]. A comparison of the PV output power for the proposed MPO and conventional tracking algorithms is depicted in Fig. 12(b). It can be observed that the proposed MPO algorithm accurately tracks the MPP under the variation of solar irradiance and temperature and provides enhanced tracking performance compared to conventional techniques. Obviously, the PV output power is decreased when temperature rises and slightly increased at lower temperature compared to the system results depicted in Fig. 11(b) wherein the temperature was fixed at 25 °C.

7. Conclusions

This paper proposed a new modified perturb and observe (MPO) MPPT algorithm for grid-integrated solar PV-systems based on an efficient open-circuit voltage estimation strategy. In order to enhance the system tracking speed with reduced contained oscillations level, the proposed algorithm divides the solar module/cell P-V curve into four sectors based on the estimated open-circuit voltage. A large perturbation step-size is applied for the two sectors located far from the MPP, which improves the tracking speed. Meanwhile, a small perturbation step-size is used for the other two sectors, which ensures minimal steady-state oscillations. Three different irradiance profiles; Ropp, sinusoidal, and one-day (10 hr.) are applied to validate the effectiveness of the proposed MPO algorithm under different varying irradiance conditions. Furthermore, the MPO algorithm tracking performance is compared with conventional LS-PO and SS-PO. The system results exhibit the proposed algorithm features and its superiority in compared with conventional techniques. The MPO algorithm reduces the number of sensors utilizing the proposed open-circuit voltage estimation technique, which reduces the system implementation cost and lowers its $/kwh. In addition, it enhances the tracking efficiency to 99.7% compared to 98.8 for the LS-PO. Moreover, it improves the tracking speed and reduces the steady-state oscillations. The proposed system performance under partially shading and dusty conditions will be considered in the future work. In addition, the impact of unbalanced grid on the proposed MPPT technique will be analysed in the future publications.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

(Tables 6-7)

Table 6
Grid-integrated PV system parameters.

System rated power, P	13.2 kW
Grid voltage, $V_{L.L}$	380 V
Grid frequency, f	50 Hz
Switching frequency, f_{SW}	5 kHz
DC bus voltage, V_{dc}	750 V
DC bus capacitance, C_{dc}	14 μF
Grid inductance and resistance, L_g, R_g	2 mH, 0.015 Ω

Table 7
GSC PI controllers gains.

Controller	k_p	k_i
Dc-link voltage	0.01	5
D-axis current	0.2	30
Q-axis current	0.2	30

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